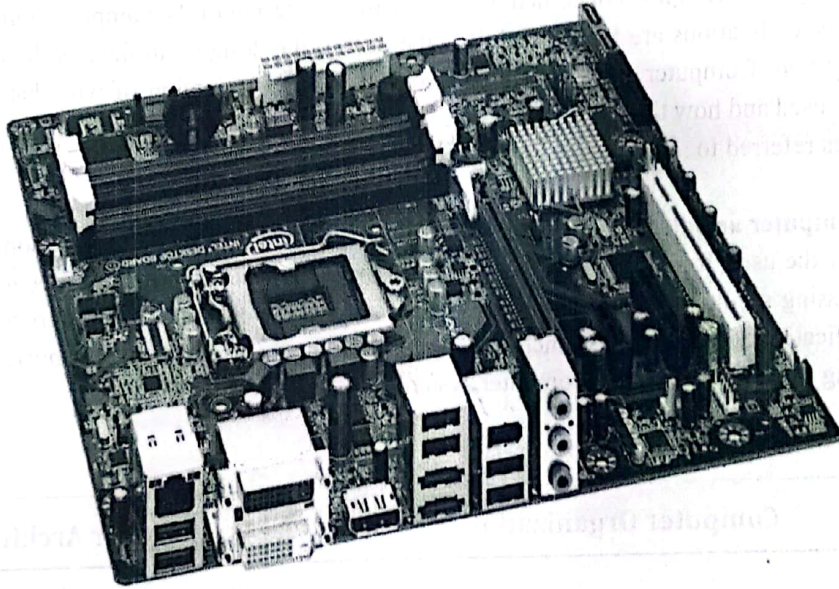


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معمارية الحاسوب، حاضرة اولي
دكتوراه، المرحلة الثالثة

Computer Architecture



Class: 3rd

Lecturer: Walaa Khashlan

Required Textbooks:

1. Harris, Sarah, and David Harris. *Digital Design and Computer Architecture: ARM Edition*. Morgan Kaufmann, 2015.
2. Tanenbaum, Andrew S. *Structured Computer Organization*. Prentice Hall, 2005.

Computer organization is concerned with the way the hardware components operate and the way they are connected together to form the computer system. The various components are assumed to be in place and the task is to investigate the organizational structure to verify that the computer parts operate as intended.

Computer design is concerned with the hardware design of the computer. Once the computer specifications are formulated, it is the task of the designer to develop hardware for the system. Computer design is concerned with the determination of what hardware should be used and how the parts should be connected. This aspect of computer hardware is sometimes referred to as computer implementation.

Computer architecture is concerned with the structure and behavior of the computer as seen by the user. It includes the information formats, the instruction set, and techniques for addressing memory. The architectural design of a computer system is concerned with the specifications of the various functional modules, such as processors and memories, and structuring them together into a computer system.

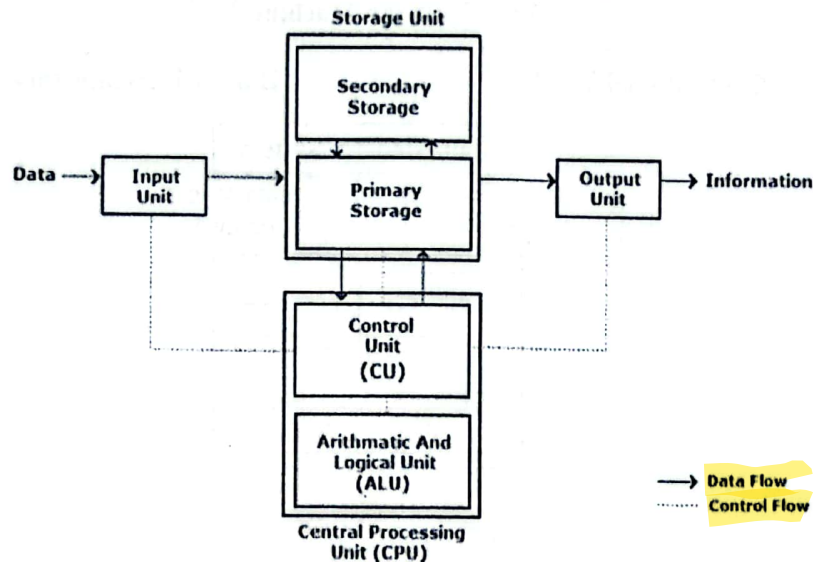
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Computer Organization	Computer Architecture
Often called microarchitecture (low level)	Computer architecture (a bit higher level)
شفاف Transparent from programmer (ex. a programmer does not worry much how addition is implemented in hardware)	Programmer view (i.e. Programmer has to be aware of which instruction set used)
Physical components (Circuit design, Adders, Signals, Peripherals)	Logic (Instruction set, Addressing modes, Data types, Cache optimization)
How to do ? (implementation of the architecture)	What to do ? (Instruction set)

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Block diagram of computer



Types of Computer Architecture

Computer architecture is divided into two types:

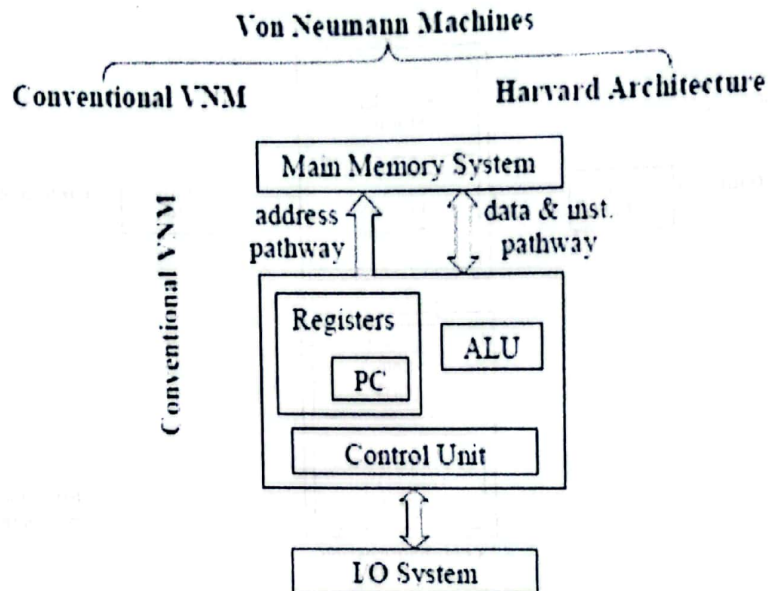
1. ISA (Instruction Set Architecture):

- Specifies the properties visible to a machine-language programmer.
- Defines the CPU functions and capabilities for arithmetic and logic, register types, memory addressing modes, and the instruction set used by programmers.

2. HAS (Hardware System Architecture):

- Specifies the efficiency and operation of the computer hardware.
- Includes the logical design and organization of the CPU, memory, and I/O parts of the computer.

Classification of Computer Architecture



1. Von Neumann Machine
2. Non-Von Neumann Machine

1. Von Neumann Machine: Proposed by the mathematician John Von Neumann in 1945 and still used today. It includes:

- **Central Processing Unit (CPU):** The electrical circuit responsible for executing computer program instructions. It includes control, arithmetic, and logic units, and registers.
- **Main Memory Unit**
- **Input/Output Units**
- **Buses** نواقل

Specifications:

- CPU, main memory, and I/O systems.
- **Stored program:** program instructions are stored in memory.

- The control unit reads instructions from memory and executes them sequentially.
- At least one pathway connects CPU and memory.

Later developments led to the Conventional Von Neumann computer:

ملخص

- CPU, main memory, and I/O.
- Stored program.
- Sequential instruction execution.
- At least two pathways between CPU and memory.

Harvard Architecture

An improvement over Von Neumann:

- Same specifications as before, but at least **four pathways** between CPU and memory for faster data transfer.

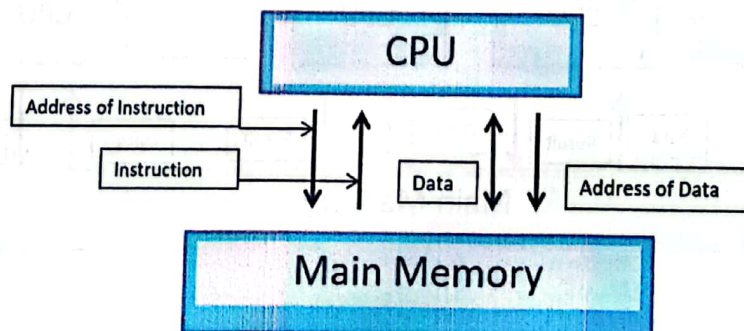
Problem with Von Neumann: it uses the same bus to fetch instructions and transfer data, creating a bottleneck.

Harvard Architecture separates program instructions and data storage and signaling.

الإشارة

In Harvard Architecture, instructions and data can be fetched simultaneously, speeding up execution. However, the control unit becomes more complex and expensive because it manages separate buses.

Modern computer architecture generally follows the Harvard model, using separate pathways for data and instructions even if the memory chip itself is unified.

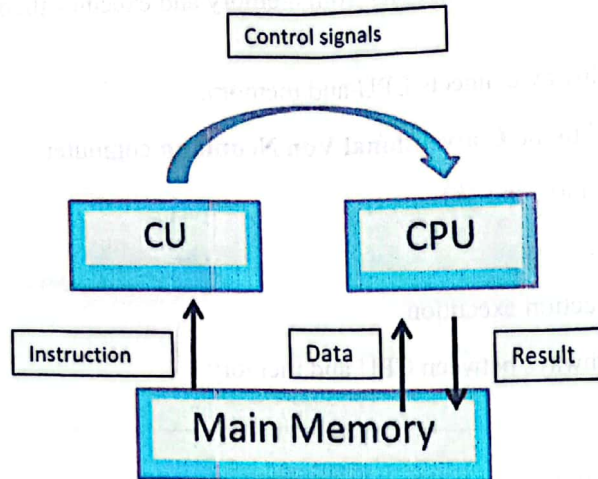


2. Non-Von Neumann Types

a. SISD (Single Instruction Stream, Single Data Stream):

- Executes one instruction on one data item at a time.
- Contains a single CPU receiving data and executing instructions in sequence.
- Used for general-purpose computing.

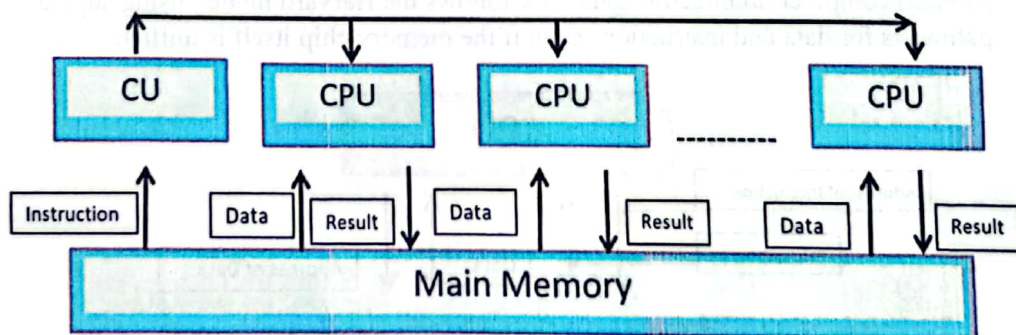
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b. SIMD (Single Instruction Stream, Multiple Data Stream):

- Executes one instruction on multiple data items simultaneously.
- Also called multi-processor computers.
- Contains several CPUs with one control unit sending control signals to all CPUs.
- Used for special purposes such as numerical analysis.

في وقت واحد



c. MISD (Multiple Instruction Stream, Single Data Stream):

- Executes multiple instructions on a single data stream.
- Rarely exists today, used for specialized complex tasks.
-

d. MIMD (Multiple Instruction Stream, Multiple Data Stream):

- Executes multiple instructions on multiple data streams.
- Multiple CPUs execute different programs on their own data simultaneously (parallel processing).
- Used in image processing and similar applications.

